

Data Analytics Internship — Executive Summary

Intern: Ritesh Kumar | Dataset: Superstore Sales (2014–2017) | Task 5 / Final Submission

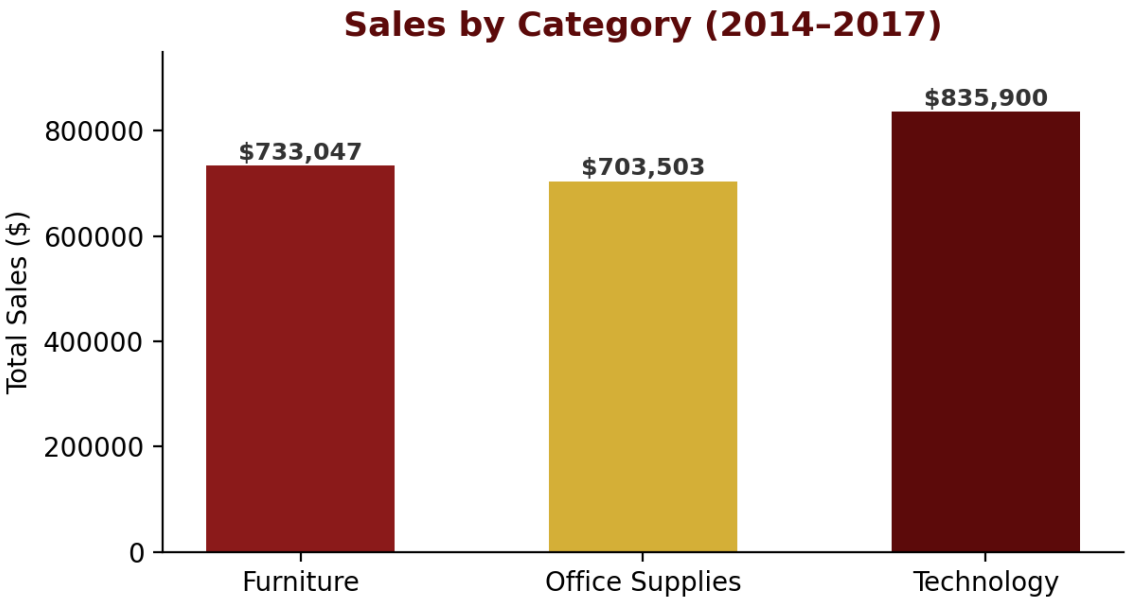
Executive Summary

This 30-day Data Analytics internship covered the full analytics lifecycle on the Superstore Sales dataset — from data cleaning and exploratory analysis, through SQL-based extraction, interactive dashboarding, to statistical testing and predictive modeling. The dataset spans 9,694 orders across four years (2014–2017), generating \$2.27M in revenue and \$282.9K in profit. Key drivers of performance were identified at the category, region, and product level, and the findings below translate directly into actionable business recommendations.

Key Business Metrics

9,694 Total Orders	\$2,272,449.85 Total Revenue	\$282,857.75 Total Profit	16% Avg. Discount
Technology Top Category	West · \$106K Top Region (Profit)	California Top State	+50.5% YoY Growth '14→'17

Dashboard Snapshot — Sales by Category



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Top 5 Key Insights

- **Technology leads category sales** at \$835,900 — ahead of Furniture (\$733,047) and Office Supplies (\$703,503) — but high-value items like the Canon imageCLASS 2200 Copier (\$61,600 alone) show revenue is concentrated in a few flagship products.
- **West and East regions drive profit** (\$106,021 and \$90,672 respectively), while Central region lags at just \$40,129 — under 40% of West's contribution despite comparable order volumes.
- **Heavy discounting erodes profitability** — orders discounted above 20% show a negative average profit (−\$9.15), while low/no-discount orders stay solidly profitable (\$68–72 avg profit).
- **Sales grew steadily year-over-year**, from \$481,764 in 2014 to \$724,995 in 2017 (+50.5%), with Standard Class remaining the dominant shipping mode by a wide margin.
- **Statistical tests found no significant regional or categorical bias** — a t-test between West and East sales ($p=0.426$) and a chi-square test of Category vs Region ($p=0.722$) both showed no significant difference, meaning regional profit gaps stem from cost/discount patterns, not skewed product mix.

Advanced Analytics Summary

Customer segmentation via K-Means clustering ($k=4$, features: Sales, Profit, Quantity, standardized) grouped customers into four distinct behavioral segments, visualized using PCA. A Linear Regression model was also built to predict Sales from Quantity, Discount, and Profit ($R^2 = -0.15$, MAE = \$245.08, RMSE = \$824.85) — the low R^2 indicates these three features alone are weak predictors of Sales, suggesting other factors (e.g., product category, customer segment) would improve future models.

Business Recommendations

- ★ **Cap discounts near 20%** for Furniture and low-margin Office Supplies sub-categories, since profit turns negative beyond this threshold — protect margins without cutting sales volume.
- ★ **Invest in the Central region** with targeted promotions or logistics improvements to close the profit gap with West/East, since no product-mix bias was found — the gap is operational, not structural.
- ★ **Double down on Technology bestsellers** (copiers, video-conferencing systems) with bundled offers, while diversifying revenue away from a handful of flagship SKUs to reduce concentration risk.